

RECKON ORB: THE ENGINEERING OPERATING SYSTEM

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I do not study isolated subjects. I build an interconnected system that helps me understand reality, solve problems, build machines, validate truth, and document progress for a lifetime.

1. Big Picture

Engineering is a **system of systems**. Every course is teaching me one part of the same machine:

- **Structures** - how things hold, deform, and fail
- **Fluids** - how things flow, accelerate, and interact with geometry
- **Thermodynamics** - how energy moves, transforms, and limits performance
- **Circuits and Control** - how systems sense, decide, and respond
- **Numerical Methods** - how difficult physics is solved computationally
- **Manufacturing** - how ideas become real parts and assemblies

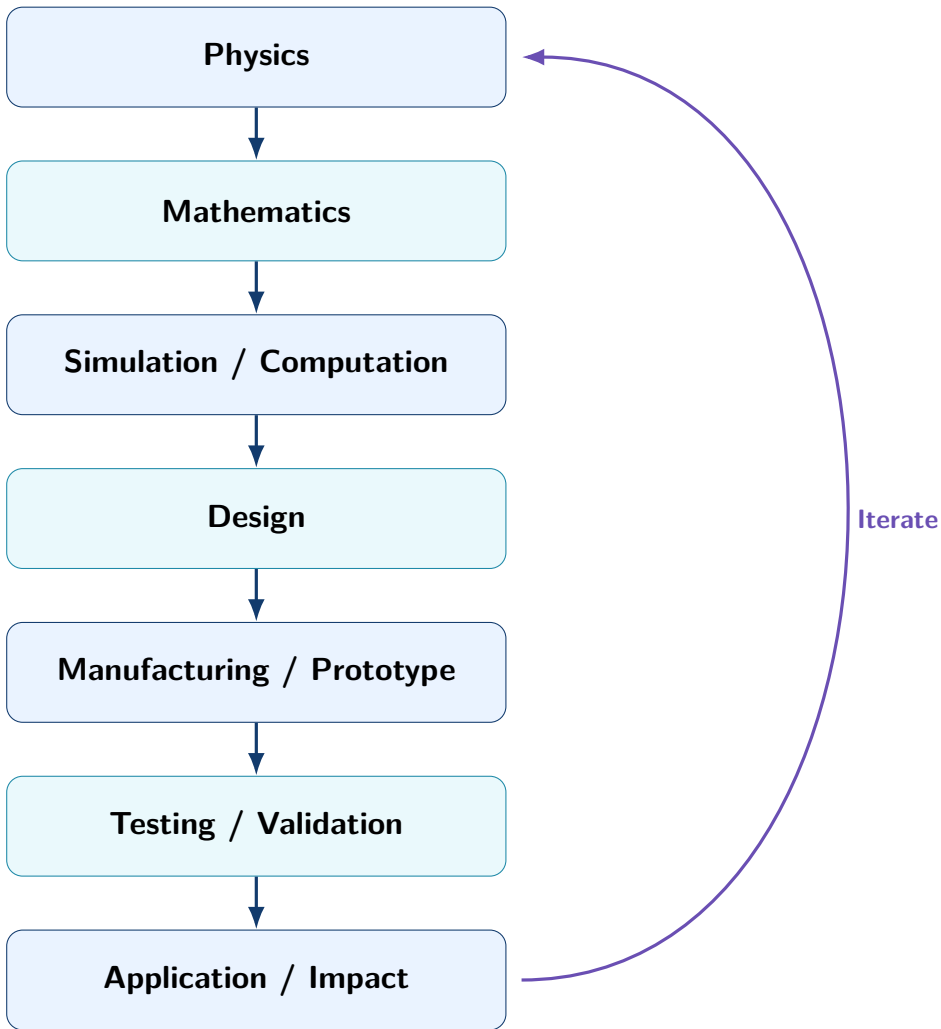
These are not random classes. Together they teach me how to **model, solve, build, test, and improve reality**.

2. Core Engineering Truth

Engineering = Model → Solve → Build → Test → Iterate

- If I cannot **model** it, I do not understand it well enough.
- If I cannot **solve** it, I cannot predict it.
- If I cannot **build** it, it is still only an idea.
- If I cannot **test** it, it is not yet trustworthy.
- If I do not **iterate** it, it will never mature.

3. The Main Pipeline

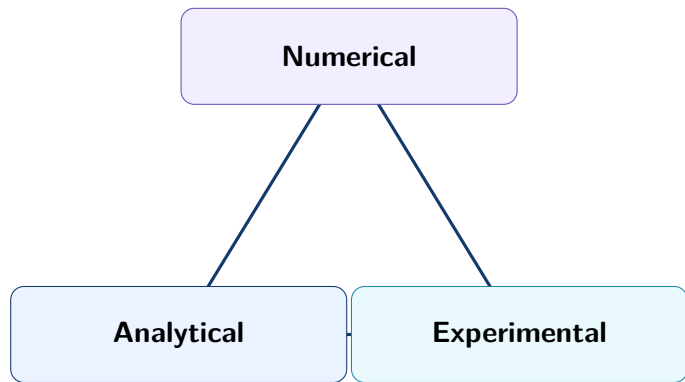


Mature engineers keep cycling through this loop with better assumptions, cleaner tools, and stronger evidence.

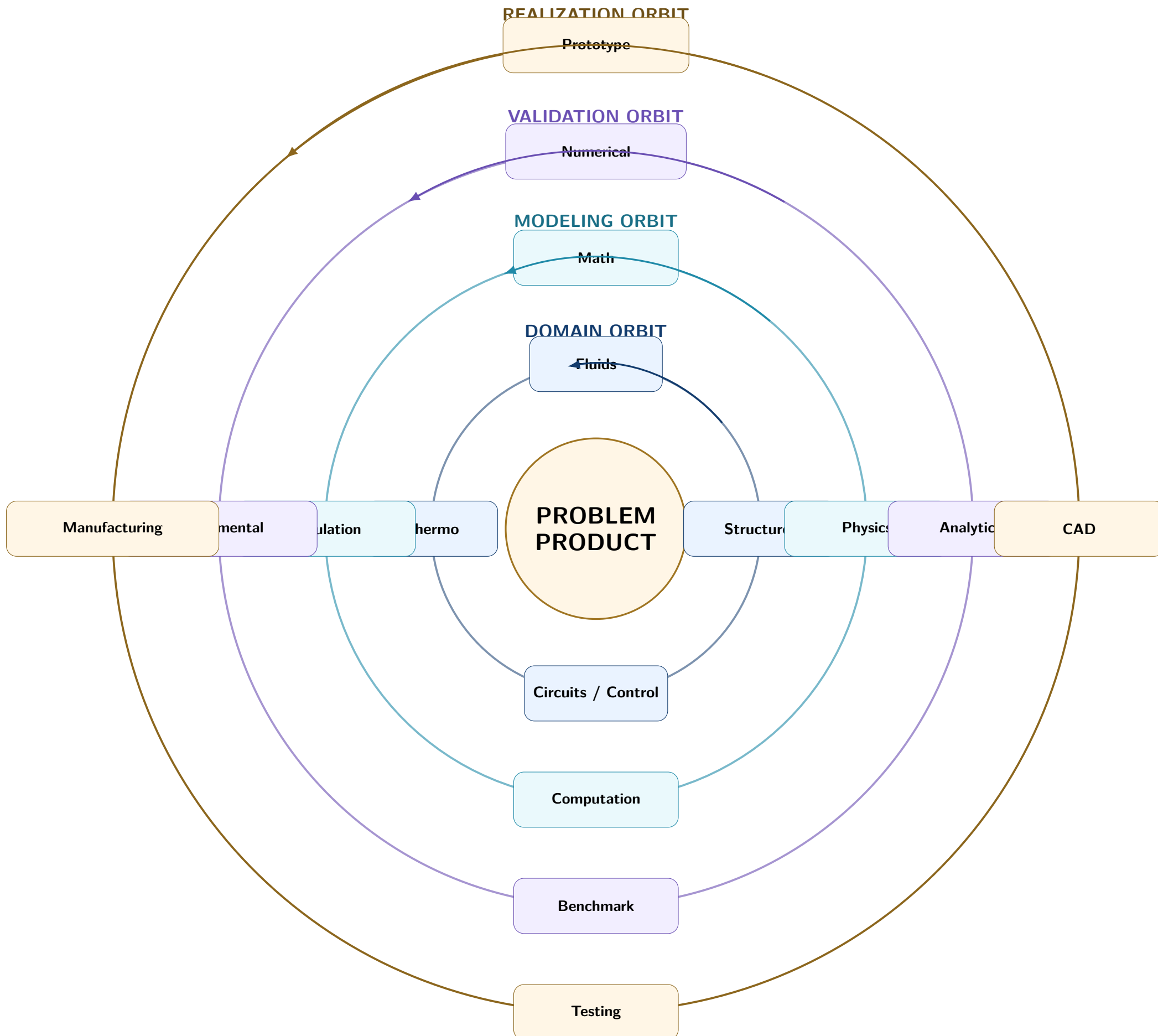
4. The Truth Triangle

Truth becomes stronger when three worlds agree:

- **Analytical** - first principles, equations, idealized reasoning
- **Numerical** - MATLAB, Python, CFD, FEA, optimization
- **Experimental** - measurements, tests, uncertainty, real data



5. Visual System Map



This is the mental picture: **everything orbits a problem**. The center is the challenge or product. The rings are the knowledge layers that let me understand it, solve it, validate it, and build it.

6. RECKON ORB and ORBIT

Reckon Orb is my system-making philosophy. Every project is an **Orb**: a mini-world revolving around a meaningful problem.

- O** Output - What am I building?
- R** Research - What must I learn?
- B** Benchmark - What does strong performance look like?
- I** Inventory - What tools, skills, data, and parts do I need?
- T** Transmission - How do I document, present, and scale it?

ORBIT turns messy ambition into repeatable progress. It forces every project to produce technical evidence, not just vague motivation.

7. Supernova Mode: Why Projects Must Connect

When multiple Orbs connect, knowledge compounds instead of resetting.

- A **rocket engine** project shares thermo, fluids, CAD, and manufacturing logic.
- A **drone** project shares controls, electronics, testing, and systems integration.
- A **re-entry** simulation shares numerics, structures, heat transfer, and modeling.

That means I am not building isolated assignments. I am building a **constellation of capability**.

8. Tool Stack

Modeling and Computation

- MATLAB
- Python
- Excel for quick sanity models

CAD and Documentation

- SolidWorks
- Technical drawings
- Assemblies and layouts

Simulation

- FEA for structures
- CFD for flow
- System-level modeling

Realization

- 3D printing
- CNC machining
- Assembly and testing

9. Formal Engineering Chain

Governing Laws → Mathematical Model → Solution Method → Implementation → Validation → Optimization This is the disciplined path from idea to reliability. It is how a concept becomes credible engineering.

10. Where This Creates Value

My long-term value does not come from memorizing formulas. It comes from becoming the kind of engineer who can:

- connect multiple disciplines,
- build and test real systems,
- document progress clearly,
- and convert knowledge into reliable technical output.

Application fields include:

- Aerospace
- Robotics
- Vehicles
- Renewable energy
- Embedded and integrated systems

11. Daily Reminder and Career Oath

I do not study subjects. I build systems.

Every project is an Orb. Every Orb follows ORBIT. Every ORBIT builds value.

Daily reminder

- If it is not built, it is not engineering.
- If it is not tested, it is not yet trustworthy.
- If it is not documented, it cannot scale.
- If it is not iterated, it will not mature.

Career oath

- Wonderful in character
- Effective in execution
- Credible in results
- Revolutionary in vision

RECKON ORB PRINCIPLE

Build. Document. Connect. Repeat.